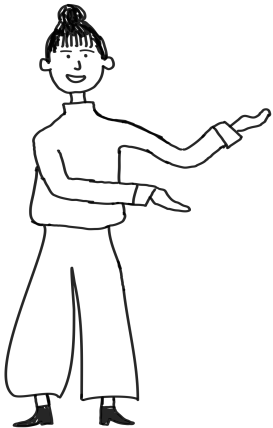
Linear Programming Basics

Lecture 2

ETF2480&ETF5248 - S2, 2026
Jessica Leung, PhD, CAP-X, SFHEA

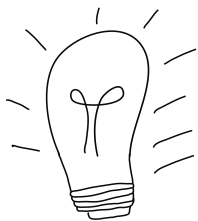
Today's Goal



- A bit more on linear algebra
- LP terminologies and transformations
- Geometric interpretation of LP
- Feasibility problem

A bit more on linear algebra

We learnt from the last lecture that we can express an LP in an elegant and compact way using matrix vector notations. Apart from looking nice, there is so much more we can do in the analysis once we know a few more tricks in linear algebra.



;

Vectors

vector of size n (or n-**vector**)

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

- we can also use the notation $x = (x_1, x_2, \ldots, x_n)$
- x_i is ith **component** or **element** (real unless specified otherwise)
- set of real n-vectors is denoted $\mathbb{R}^n$

special vectors (with n determined from context)

- $x = 0$ (zero vector): $x_i = 0$, $i = 1, \ldots, n$
- $x = 1$ (vector of ones): $x_i = 1$, $i = 1, \ldots, n$
- $x = e_i$ (ith basis or unit vector): $x_i = 1$, $x_k = 0$ for $k \neq i$

Vector Operations

Scalar vector multiplication: When a vector is multiplied by a scalar (i.e., number), we multiply every element of the vector by the scalar.

$$2\begin{bmatrix}1\\-2\\3\end{bmatrix} = \begin{bmatrix}2\\-4\\6\end{bmatrix}$$

Vector Operations

Addition: Two vectors of the same size can be added together by adding the corresponding elements, to form another vector of the same size.

$$\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} + \begin{bmatrix} 8 \\ -1 \\ 9 \end{bmatrix} = \begin{bmatrix} 9 \\ -3 \\ 12 \end{bmatrix}$$

The standard **inner product** on R^n, the set of real n-vectors, is given by

$$x^T y = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n = \sum_{i=1}^{n} x_i y_i$$

for $x, y \in \mathrm{R}^n$.

Matrices

matrix of size $m \times n$:

$$A = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{bmatrix}$$

- A_{ij} (or a_{ij}) is ijth **element** (**entry** or **coefficient**)
- set of $m \times n$ matrices is denoted $\mathrm{R}^{m \times n}$
- vectors can be viewed as matrices with one column

special matrices (with size determined from context)

- $X = 0$ (zero matrix): $X_{ij} = 0$, $i = 1, \ldots, m$, $j = 1, \ldots, n$
- $X = I$ (identity matrix): $m = n$ with $X_{ii} = 1$, $X_{ij} = 0$ for $i \neq j$

Matrix operations

Scalar multiple: given any matrix A and any number c, the matrix cA is obtained from the matrix A by multiplying each element of A by c.

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}; \quad 3A = \begin{bmatrix} 3 & 6 \\ -3 & 0 \end{bmatrix}.$$

Addition: let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices with the same order. Then $C = A + B$ is defined to be the $m \times n$ matrix whose ij-th element is $a_{ij} + b_{ij}$.

$$A = \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 3 & -1 \end{bmatrix}; \quad B = \begin{bmatrix} -1 & 2 \\ -2 & 1 \\ -3 & -1 \end{bmatrix}; \quad C = A + B = \begin{bmatrix} 0 & 2 \\ 0 & 0 \\ 0 & -2 \end{bmatrix}.$$

Transposition

Transposition: given any $m \times n$ matrix A, the transpose of A (written A^T) is the $n \times m$ matrix given by $(A^T)_{ij} = a_{ji}$.

For example:

$$A = \begin{bmatrix} 0 & 4 \\ 7 & 0 \\ 3 & 1 \end{bmatrix}; \quad A^T = \begin{bmatrix} 0 & 7 & 3 \\ 4 & 0 & 1 \end{bmatrix}.$$

The same transposition is used in inner products of vectors as well.

Matrices and systems of linear equations

Consider a system of m linear equations with n variables

$$\begin{array}{rcccccl} A_{11}x_1 & + & A_{12}x_2 & + \cdots + & A_{1n}x_n & = & b_1 \\ A_{21}x_1 & + & A_{22}x_2 & + \cdots + & A_{2n}x_n & = & b_2 \\ & & & & & \vdots & \\ A_{m1}x_1 & + & A_{m2}x_2 & + \cdots + & A_{mn}x_n & = & b_m \end{array}$$

These equations can be written in matrix notation as

$$Ax = b$$

where

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Expressing constraint $x \geq 0$ within $Ax \geq 0$.

Expressing constraint $x \geq 0$ within $Ax \geq 0$.

Does this change our coding syntax?

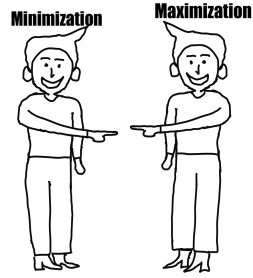
For those who want to learn **more** about these notations beyond this unit...

A very easy to read (free) textbook, designed for first year students:
Boyd and Vandenberghe (2018) *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*
Read Ch. 1.1-1.4 (vectors), 6.1-6.4(matrix), 10.1(matrix multiplication)
Link: `https://web.stanford.edu/~boyd/vmls/vmls.pdf`

For Python implementation of Boyd and Vandenberghe (2018):
Leung and Matsypura (2019) *Python Language Companion to Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*
Link: `https://hdl.handle.net/2123/21370`

LP Terminologies and Useful Transformations

We need to know what we are referring to in an LP before we dive deeper.



Linear programming (LP) problem

$$\begin{aligned} \text{minimize} \quad & \sum_{j=1}^{n} c_j x_j \\ \text{subject to} \quad & \sum_{j=1}^{n} a_{ij} x_j \leq b_i, \quad i = 1, \ldots, m \\ & \sum_{j=1}^{n} d_{ij} x_j = f_i, \quad i = 1, \ldots, p \end{aligned}$$

in matrix notation:

$$\begin{aligned} \text{minimize} \quad & c^T x \\ \text{subject to} \quad & Ax \leq b \\ & Dx = f \end{aligned}$$

- A is $m \times n$-matrix with elements a_{ij}, rows a_i^T
- D is $p \times n$-matrix with elements d_{ij}, rows d_i^T
- inequality is component-wise vector inequality

Terminology

$$\begin{array}{rl} \text{minimize} & c^T x \\ \text{subject to} & Ax \leq b \\ & Dx = f \end{array}$$

- x is **feasible** if it satisfies the constraints $Ax \leq b$ and $Dx = f$
- **feasible set** is set of all feasible points
- $x^\star$ is **optimal** if it is feasible and $c^T x^\star \leq c^T x$ for all feasible x
- the **optimal value** of the LP is $p^\star = c^T x^\star$
- **unbounded problem:** $c^T x$ unbounded below on feasible set ($p^\star = -\infty$)
- **infeasible problem:** feasible set is empty ($p^\star = +\infty$)

Maximisation problems

We can solve the **maximisation problem**

$$\begin{array}{rl} \text{maximize} & c^T x \\ \text{subject to} & Ax \leq b \\ & Dx = f \end{array}$$

by minimising the function $-c^T x$ subject to the same constraints:

$$\begin{array}{rl} \text{minimize} & -c^T x \\ \text{subject to} & Ax \leq b \\ & Dx = f \end{array}$$

with the following changes in terminology:

- $x^\star$ is **optimal** if it is feasible and $c^T x^\star \geq c^T x$ for all feasible x
- **unbounded problem:** $c^T x$ unbounded above on feasible set ($p^\star = +\infty$)
- **infeasible problem:** feasible set is empty ($p^\star = -\infty$)

Geometric Interpretation of LP

We can visualise the process of solving an LP
on a graph easily when we only have 2 decision variables.



Giapetto's Woodcarving

Giapetto's Woodcarving, Inc., manufactures wooden soldiers and trains.
Each soldier built:

- Sells for $27 and uses $10 worth of raw materials.
- Increases Giapetto's variable labor/overhead costs by $14.
- Requires 2 hours of finishing labor.
- Requires 1 hour of carpentry labor.

Each train built:

- Sells for $21 and uses $9 worth of raw materials.
- Increases Giapetto's variable labor/overhead costs by $10.
- Requires 1 hour of finishing labor.
- Requires 1 hour of carpentry labor.

Each week Giapetto can obtain:

- All needed raw material.
- Only 100 finishing hours.
- Only 80 carpentry hours.

Demand for the trains is unlimited. At most 40 soldiers are bought each week. Giapetto wants to maximize weekly profit. Formulate and solve an LP.

Giapetto's Woodcarving: LP formulation

$$
\begin{array}{rlrl}
\text{maximize} & 3x_1 + 2x_2 & & \\
\text{subject to} & 2x_1 + x_2 & \leq \quad 100 & \text{(finishing constraint)} \\
& x_1 + x_2 & \leq \quad 80 & \text{(carpentry constraint)} \\
& x_1 & \leq \quad 40 & \text{(constraint on demand for soldiers)} \\
& x_1 & \geq \quad 0 & \\
& x_2 & \geq \quad 0 &
\end{array}
$$

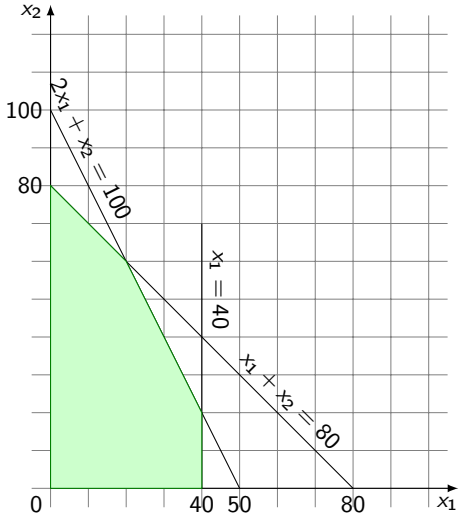
in matrix notation:

$$
\begin{array}{rl}
\text{maximize} & c^T x \\
\text{subject to} & Ax \leq b
\end{array}
$$

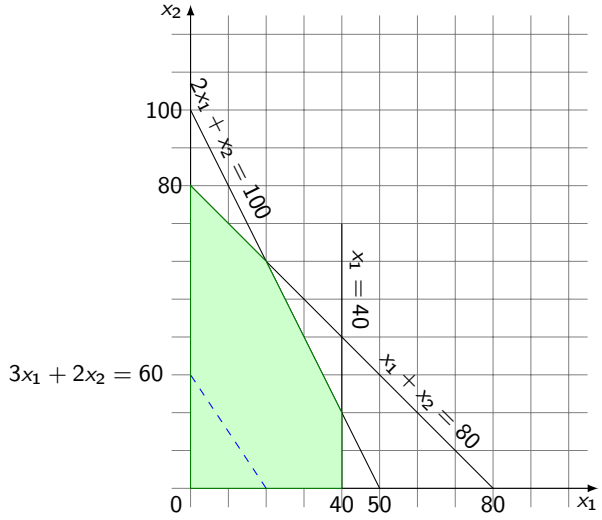
where

$$
A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \\ 1 & 0 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}, \quad b = \begin{bmatrix} 100 \\ 80 \\ 40 \\ 0 \\ 0 \end{bmatrix}, \quad c = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.
$$

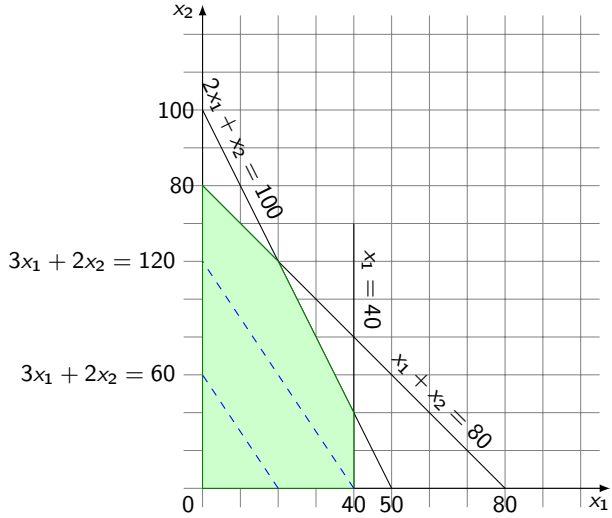
Giapetto's Woodcarving: geometric interpretation



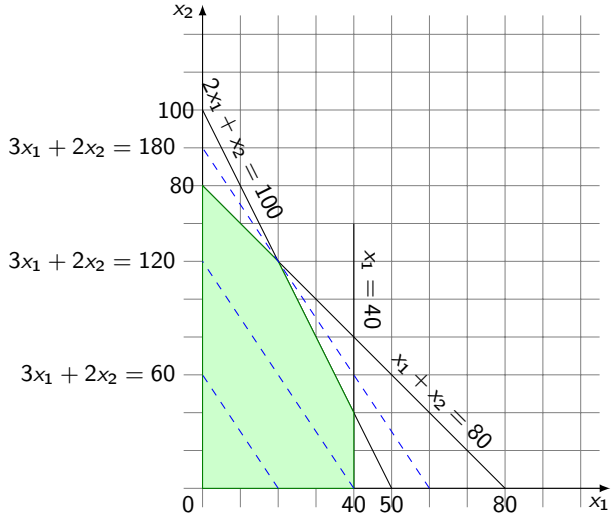
Giapetto's Woodcarving: geometric interpretation



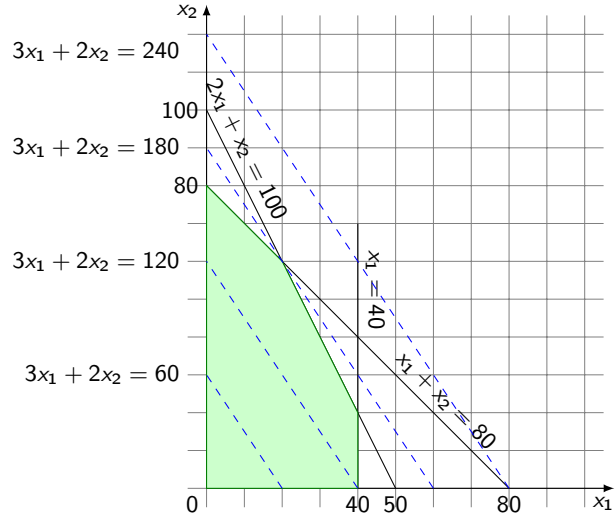
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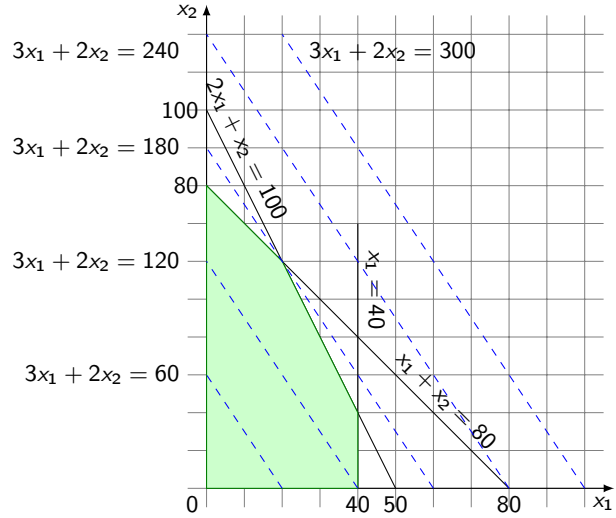
Giapetto's Woodcarving: geometric interpretation



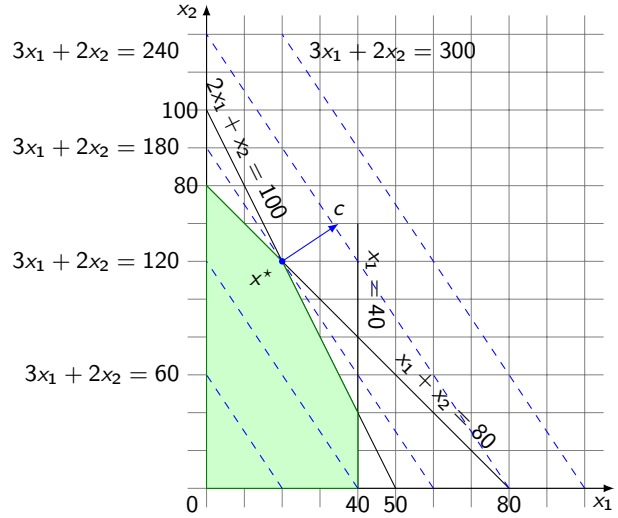
Giapetto's Woodcarving: geometric interpretation



Giapetto's Woodcarving: geometric interpretation



Giapetto's Woodcarving: geometric interpretation



optimal solution is $x^\star = (20, 60)$ with $p^\star = 180$

Live Coding Time!

Let's code the Giapetto's Woodcarving problem together! We can also look at how the objective function moves across the polyhedron.

Multiple solutions, infeasibility and unboundedness

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax \geq b \end{array}$$

- **unbounded problem:** $c^T x$ unbounded below on feasible set ($p^\star = -\infty$)
- **infeasible problem:** feasible set is empty ($p^\star = +\infty$)
- **multiple (alternative) solutions:** $c^T x$ is parallel to one of the constraints

In fact, every LP must fall into one of the following four cases:

- **Case 1:** LP is infeasible
- **Case 2:** LP is unbounded
- **Case 3:** LP has a unique solution
- **Case 4:** LP has multiple solutions

Feasibility Problem

Sometimes we don't really have something to maximize/minimize, all we care is if there is a solution that satisfies all the constraints or not.



(Linear) feasibility problem

$$\begin{array}{rl} \text{find} & x \\ \text{subject to} & Ax \geq b \end{array}$$

can be considered a special case of the general LP problem with $c = 0$:

$$\begin{array}{rl} \text{minimize} & 0 \\ \text{subject to} & Ax \geq b \end{array}$$

- $p^\star = 0$ if constraints are feasible; any feasible x is optimal
- $p^\star = +\infty$ if constraints are infeasible

Quick Summary

Today we have learnt

- a little bit more about linear algebra and terminologies that helps us learn LP better
- we can transform the LPs so that they look similar to one another
- some LP can be solved graphically
- sometimes we only want to see if it's feasible



Appendix: Hyperplanes, halfspaces and polyhedrons

Hyperplanes and halfspaces

Definition: hyperplane is a solution set of one linear equation with nonzero coefficient vector a

$$a^T x = b$$

Definition: halfspace is a solution set of one linear inequality with nonzero coefficient vector a

$$a^T x \geq b$$

a is the **normal vector**

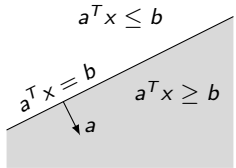


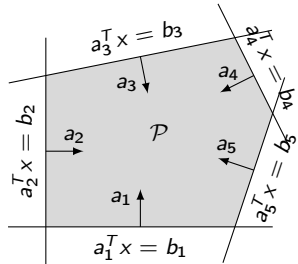
Figure: A hyperplane and two halfspaces.

Polyhedron

Definition: polyhedron is the solution set of a finite number of linear inequalities:

$$a_1^T x \geq b_1, \quad a_2^T x \geq b_2, \quad \ldots, \quad a_m^T x \geq b_m$$

- intersection of a finite number of halfspaces
- in matrix notation: $Ax \geq b$ if A is a matrix with rows a_i^T
- can include equalities: $Fx = g$ is equivalent to $Fx \geq g$, $-Fx \geq -g$



Feasible set

Feasible set (or **feasible region**, **constraint set**) for an LP is the set of all points that satisfy all the constraints of the LP.

Property: the feasible set of an LP is a **polyhedron**.

Proof: A polyhedron is defined as the solution set of a finite number of linear inequalities:

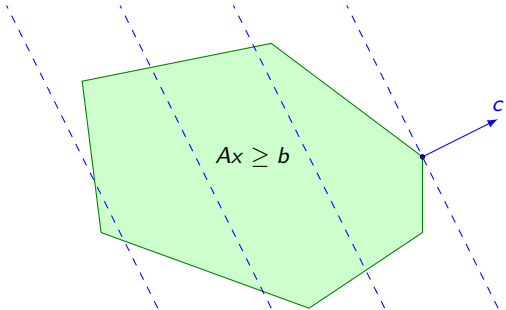
$$\mathcal{P} = \{x \mid a_i^T x \geq b_i, i = 1, \ldots, m\}$$

Because in an LP, each constraint must be either a linear equation or linear inequality and there is a finite number of them, **feasible set** is a polyhedron. ■

Geometric interpretation of LP

maximize $c^T x$

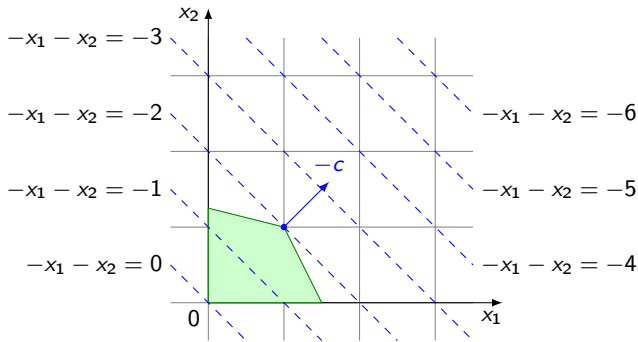
subject to $Ax \geq b$



dashed lines (hyperplanes) are level sets $c^T x = \alpha$ for different α

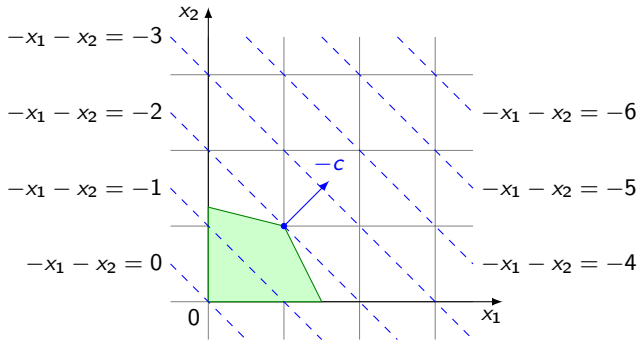
Workshop Example

$$
\begin{aligned}
\text{minimize} \quad & -x_1 - x_2 \\
\text{subject to} \quad & 2x_1 + x_2 \leq 3 \\
& x_1 + 4x_2 \leq 5 \\
& x_1 \geq 0, \quad x_2 \geq 0
\end{aligned}
$$



Workshop Example

$$
\begin{aligned}
\text{minimize} \quad & -x_1 - x_2 \\
\text{subject to} \quad & 2x_1 + x_2 \leq 3 \\
& x_1 + 4x_2 \leq 5 \\
& x_1 \geq 0, \quad x_2 \geq 0
\end{aligned}
$$



optimal solution is $x^\star = (1, 1)$ with $p^\star = -2$

Workshop Example

- What would the visualization look like in a 2D example if the objective function is no longer a linear function?
- How would the feasible region change if the constraints are not linear?
- Can we generate an interactive graphic with the help of GenAI?

(If time permits) Product Mix (Marketing Application)

Gurobi is a state-of-the-art commercial solver to solve optimization problem. The company's target customers are academics (professors and students) and industry practitioners. To reach these groups, Gurobi is considering to purchase full page advertisement in two magazines: "Analytics Today" and "Analytics Tomorrow".

- An "Analytics Today" ad (in each issue) is seen by 7 million academics and 2 million industry practitioners and costs $50,000.
- An "Analytics Tomorrow" ad (in each issue) is seen by 2 million academics and 12 million industry practitioners and costs $100,000.
- Gurobi would like for commercials to be seen by at least 28 million academics and 24 million industry practitioners.

Use LP to determine how Gurobi can meet its advertising requirements at minimum cost.

(If time permits) Product Mix (Marketing Application): solution

Gurobi must decide how many "Analytics Today" and "Analytics Tomorrow" ads should be purchased, so the decision variables are

- x_1 - number of "Analytics Today" ads purchased
- x_2 - number of "Analytics Tomorrow" ads purchased

Then the problem can be formulated as:

$$\begin{aligned} \text{minimize} \quad & 50x_1 + 100x_2 \\ \text{subject to} \quad & 7x_1 + 2x_2 \geq 28 \\ & 2x_1 + 12x_2 \geq 24 \\ & x_1, x_2 \geq 0 \end{aligned}$$